LP3 Assignment A3

Roll no: 41168

Problem Statement:
Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy.

```
#include <bits/stdc++.h>
using namespace std;

struct Item { int w, v; };

int main() {
    int n, W;
    cout << "Enter number of items and capacity: ";
    cin >> n >> W;
    vector<Item> a(n);
    cout << "Enter weight and value of each item:\n";
    for (auto& it : a) cin >> it.w >> it.v;

    vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));
    for (int i = 1; i <= n; i++) {
        for (int c = 0; c <= W; c++) {
            dp[i][c] = dp[i - 1][c];
            if (a[i - 1].w <= c)
                dp[i][c] = max(dp[i][c], dp[i - 1][c - a[i - 1].w] + a[i - 1].v);
        }
    }
    cout << "Max value: " << dp[n][W] << "\nSelected items (weight, value):\n";

    for (int i = n, c = W; i > 0; i--)
        if (dp[i][c] != dp[i - 1][c]) {
            cout << "Item " << i << ": (" << a[i - 1].w << ", " << a[i - 1].v << ")\n";
            c -= a[i - 1].w;
        }
}
```

OUTPUT:

```
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ g++ LP3A3.cpp -o a
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ ./a
Enter number of items and capacity: 5
40
Enter weight and value of each item:
4
2
6
3
1
4
2
5
6
3
Max value: 17
Selected items (weight, value):
Item 5: (6, 3)
Item 4: (2, 5)
Item 3: (1, 4)
Item 2: (6, 3)
Item 1: (4, 2)
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
```